

① Git

① Version control system is a tool that helps to track changes in code.

It is popular

free & open source.

fast and scalable.

② GitHub

① Website that allows developers to store and manage their code using Git.

1. `git -- version`

2. `clone` - cloning a repository on our local machine.
`git clone (link)`

3. `cd` → change directory.

4. `ls` → list of files.

5. `ls -a` → all files.

6. `git status` → display the status of the code.

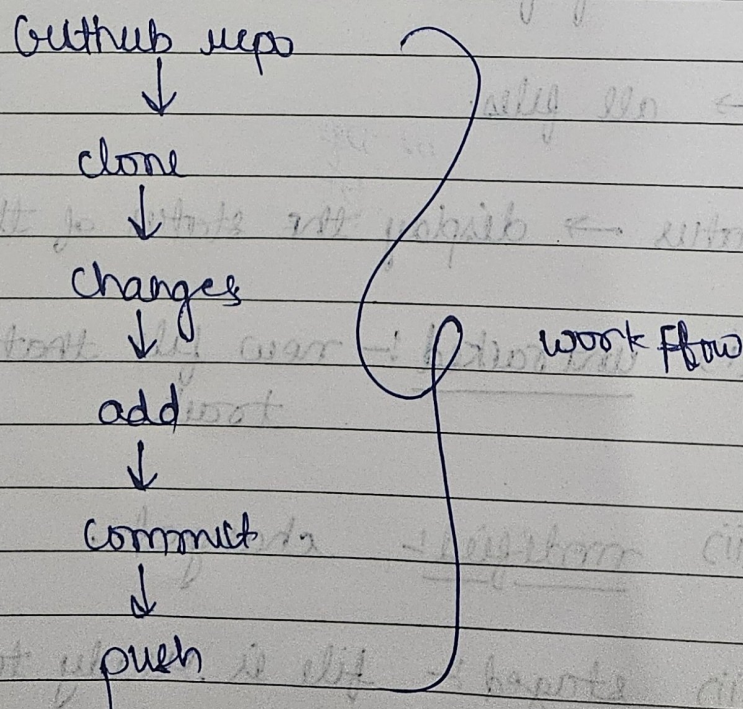
(i) untracked :- new file that git doesn't yet track

(ii) modified :- changed.

(iii) staged :- file is ready to be committed

7. **add:-** add new and changed files in your working directory to the Git staging area.
`git add <file name>.`
8. **commit:-** it is the record of change.
`git commit -m "some message"`
9. **push:-** upload local repo to remote repo.
`git push origin main`
10. **init:-** used to create a new git repo.
`git init`
`git remote add origin <url>`
`git remote -v` (to verify remote).
`git branch` (to check branch).
`git branch -M main` (to rename branch).
`git push origin main`

→ Work Flow



① Git Branches

1. `git branch`
2. `git branch -m <new name>` to change name.
3. `git checkout -b <branch name>` to create new branch.
4. `git branch -d <branch name>` to delete branch
5. `git push origin <branch name>` to push changes.
6. `git diff <branch name>` to compare two branches
7. `git merge <branch name>` to merge two branches

② pull request

It lets you tell others about changes you've pushed to a branch in a repository on GitHub.

③ Resolving merge conflicts:-

- ① An event that takes place when git is unable to automatically resolve differences in code between two commits.

④ undoing changes:-

1. (Case 1): staged changes
`git reset -- filename->`
`git reset`

2. Case 2:- Commit changes (for one commit)
git reset HEAD~1.

3. Case 3:- Committed changes (for many commits)
git reset <- commit hash ->
git reset -- hard <- commit hash ->.

① Fork:-

② A fork is a new repository that shares code and visibility settings with the original "upstream" repository.

→ Fork is a rough copy.